

TEST-07-COMMUNICATIONS-TESTING

TEST-07: PROTONIC COMMUNICATIONS TESTING PROTOCOL

Carrington Storm Motors — Project AEGIS

H-FET Transceiver • Faraday Alarm • Protonic Backup Channel • Range • EMI Immunity • Redundancy

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Reference Standards: FCC Part 15 (petitioned exemption), IEEE 299 (EMI testing), IEC 61000-4-x (immunity)

PROTONIC TESTING PHILOSOPHY

The Protonic Communication Network is the first non-electromagnetic communication system in human history. Testing it requires acknowledging that standard RF test equipment (spectrum analyzers, signal generators, network analyzers) was designed for electron-based systems. Where standard equipment is inadequate, CSM has developed custom test fixtures for protonic signal generation, detection, and characterization. These fixtures themselves become the reference standard for the emerging field of protonic communications metrology.

PART 1: H-FET TRANSISTOR CHARACTERIZATION (TRL 3→4)

1.1 Individual H-FET Test Protocol

Parameter	Method	Equipment	Target
Transfer Characteristic (Id vs. Vgs)	Sweep Vgs from 0 V to 3 V, measure Id; Nafion 115 membrane, 25°C, 100% RH	Source meter (Keysight B2902A), environmental chamber for humidity control	On/off ratio >10 ⁵ at Vgs=1.8 V; subthreshold swing <200 mV/decade
Output Characteristic (Id vs. Vds)	Sweep Vds from 0 V to 2 V at fixed Vgs=1.8 V; measure Id	Same as above	Output conductance (dId/dVds) in saturation <1 μS
Gate Leakage Current	Measure current through MXene gate electrode at Vgs=1.8 V	Source meter, 1 pA resolution	<10 pA (confirms MXene gate is not shorted to channel through pinholes in Nafion)
Proton Transport Number	Measure hydrogen gas evolution at cathode (Faradaic efficiency)	Gas chromatography or mass flow meter at cathode	>0.98 (98% of current is proton transport, <2% is electron leakage through Nafion)
Frequency Response	Apply small-signal AC gate voltage (10 mV RMS) + DC bias; measure drain current amplitude and phase	Lock-in amplifier (SR830) + source meter	Cutoff frequency (fT) >100 Hz (sufficient for 50 baud text and 3.4 kHz voice; not competing with GHz CMOS)
	Repeat transfer	Environmental chamber +	On/off ratio >10 ⁴ at 0°C (proton conductivity decreases with temperature — Nafion)

Temperature Dependence	characteristic at 0°C, 25°C, 60°C, 85°C	source meter	requires >0°C); ratio >10 ⁵ at 60°C (optimal); ratio >10 ³ at 85°C (membrane dehydration — requires active humidification)
Lifetime	Continuous operation at V _{gs} =1.8 V, 25°C, 100% RH	Automated test station logging Id every minute	Target >50,000 hours (6 years continuous); degradations modes: gate MXene oxidation, Nafion sulfonic acid group loss, membrane mechanical fatigue
MXene Gate Stability	XPS before/after 1,000 hours of operation	XPS (surface termination chemistry)	F-termination <5 at% maintained; O-termination >60 at% maintained; no Ti oxidation (TiO ₂ peak absent)

1.2 Logic Gate Family Validation

Gate Type	Configuration	Test	Pass/Fail
NOT (Inverter)	Single H-FET with resistive load	Input 0/1.8 V; measure output	Output = NOT(input) to within 10% of rail
NAND	2 H-FETs in series	Truth table (00, 01, 10, 11)	All 4 states correct
NOR	2 H-FETs in parallel	Truth table	All 4 states correct
AND	NAND + NOT in series	Truth table	All 4 states correct
OR	NOR + NOT in series	Truth table	All 4 states correct
Ring Oscillator	5-stage ring oscillator (odd number of NOT gates)	Measure oscillation frequency	Oscillation confirmed (proves gain >1 per stage); frequency stable within ±10% over 1 hour
Yield	Fabricate 100 transistors per wafer; test all	≥60% functional (production target)	

Cost: \$320K; Duration: 4 months

PART 2: PROTONIC TRANSCEIVER — INTEGRATED PROTOTYPE (TRL 4→5→6)

2.1 Transceiver Specifications

Parameter	Target	Measurement Method
Voice Bandwidth	300 Hz–3.4 kHz (telecom-grade voice)	Frequency response measurement (sweep input frequency, measure output amplitude)
Data Rate — Text	50 baud	BER test with 10 ⁶ bit pseudo-random sequence
Data Rate — Sensor	100 baud	BER test
Output Power (Protonic)	Not applicable (no RF power) — instead measure proton flux at receiver	Custom protonic flux detector (Pd-based hydrogen sensor or mass spectrometer at receiver membrane)
Range	10 m (lab, Nafion-only); 1 km target (field, Nafion + PMMA POF backbone)	Graduated distance test
Latency	<2.8 μs/mm through Nafion (proton drift velocity) + optical delay in POF backbone	Time-domain reflectometry or pulse measurement

Power Consumption	<500 mW (TX+RX combined, excluding POF transceivers)	DC power supply measurement
Self-Powered Operation	100% powered by $\text{KNbO}_3\text{-BaTiO}_3$ piezoelectric harvester (ambient vibration)	Operate transceiver for 24 hours with only piezo harvester input (no DC supply); vibration simulates walking or machinery mounting

2.2 EMI Immunity Testing — The Critical Differentiator

Standard communication equipment is tested for EMI immunity (IEC 61000-4-3, 10 V/m). The protonic transceiver is tested at Carrington-equivalent levels:

Test	Level	Standard	Performance Criterion
Radiated RF (IEC 61000-4-3)	100 V/m (10× standard level)	80 MHz–6 GHz	Criterion A (no degradation)
Magnetic Field (IEC 61000-4-8)	1,000 A/m continuous (33× standard 30 A/m)	50 Hz and 60 Hz	Criterion A — PROTON MASS CONFIRMED IMMUNE
Pulse Magnetic Field (IEC 61000-4-9)	1,000 A/m peak (10× standard)	8/20 μs pulse	Criterion A
Damped Oscillatory Magnetic (IEC 61000-4-10)	100 A/m (standard level)	100 kHz and 1 MHz	Criterion A
Power Frequency Magnetic (special test)	1,000 A/m at 0.001–1 Hz (geomagnetic band)	Custom Carrington simulation	Criterion A — THIS IS THE KILLER TEST FOR CONVENTIONAL RADIOS (which fail completely due to antenna coupling). The protonic transceiver passes because it has no antenna.

Expected Result: Protonic transceiver shows zero measurable degradation at any EMI level — because proton mass = 1,836 × electron mass, and the Lorentz force on a moving proton in a changing magnetic field is 1,836 times smaller than on an electron. The transceiver’s immunity is not from shielding — it is from physics.

PART 3: FARADAY ALARM TRIGGER SENSITIVITY

3.1 Product Description

The Faraday Alarm is a consumer/industrial device that triggers an alert when the local magnetic field change rate (dB/dt) exceeds a threshold indicative of an impending Carrington Event. It provides 15-60 minutes of warning (CME transit time from L1 Lagrange point to Earth after satellite detection).

3.2 Test Protocol

Parameter	Method	Target
dB/dt Sensitivity	Apply controlled dB/dt via Helmholtz coil; sweep from 1 nT/s to 10,000 nT/s	Alarm triggers at user-settable threshold (default: 500 nT/min — well below Carrington 2,000 nT/min, providing early warning)
False Alarm Rate	Operate for 30 days in ambient magnetic environment (including nearby appliances, vehicles, elevators)	<1 false alarm per 30 days
Miss Rate	Apply 100 simulated CME events (dB/dt waveform from historical Carrington, 1921, 1989, 2003, 2024 storms)	100% detection rate (zero missed events)
Alert Latency	Measure time from dB/dt crossing threshold to alarm output (visual +	<100 ms

Power	audible + network message) 100% piezo-powered (no battery); verify alarm functions with ambient vibration only	Operational within 1 second of vibration onset
Communication	Upon trigger, send alert via LoRa mesh (LandOLil network) and/or Protonic Network	Message received at base station within 5 seconds of trigger
EMI Immunity	Operate alarm during 1,000 A/m magnetic field test (same as protonic transceiver) — alarm must not false- trigger from the test field (must discriminate between natural dB/dt and test field)	No false trigger during test; correct trigger after test when simulated CME waveform applied

PART 4: PROTONIC BACKUP CHANNEL RELIABILITY

4.1 Redundancy Architecture

The Protonic Network is designed as the backup communication channel for critical infrastructure — it activates when primary RF/satellite/cellular channels fail. Testing must verify this failover behavior.

4.2 Failover Test Protocol

Test Scenario	Primary Channel	Protonic Backup	Acceptance
RF Jamming (simulated)	UHF/VHF radio	Protonic auto-detects primary failure (no heartbeat for 30 seconds) → activates protonic channel → restores communication	Communication restored within 60 seconds of primary failure
GPS Denial	GPS for timing/navigation	Protonic channel provides alternative timing (1 pulse per second from local oscillator synchronized via protonic network time protocol)	Timing accuracy <1 ms drift over 24 hours without GPS
Cellular Outage	4G/5G modem	Protonic + LoRa mesh provides SMS-equivalent text messaging	Message delivered within 30 seconds of send; >95% delivery rate
Satellite Outage	Iridium/Inmarsat terminal	Protonic + PMMA POF backbone carries equivalent low-bandwidth data	Data delivered with <5% packet loss
Power Grid Failure	All systems on mains power	Protonic transceiver self- powered (piezo); PMMA POF backbone powered by LiFePO ₄ batteries (48 hour runtime)	All protonic nodes remain operational throughout grid outage

PART 5: PROTONIC NETWORK — FIELD DEPLOYMENT TESTING (5-Node Pilot)

5.1 Test Network Configuration (from ROAD-06, Pilot 5)

Node	Location	Test Metrics
Node 1 — Master	CSM El Segundo	Reference for all measurements; continuous calibration source
Node 2 — Substation	Pacific NW substation (Infrastructure Pilot)	GIC environment; EM noise floor from 60 Hz power equipment

Node 3 — Rural	Cascadia community center	Standalone operation (no grid, no cellular); long-term piezo power sustainability
Node 4 — Maritime	Charlemagne Pilot Vessel	Mobile node; saltwater environment; vessel motion (antenna tracking for POF link)
Node 5 — EOC	FEMA Region X emergency operations center	Interoperability with government emergency communication systems

5.2 12-Month Data Collection

Metric	Target	Measurement
Network Uptime	>99.5%	Heartbeat monitoring at 1-minute intervals
Voice Quality (MOS)	>3.5	Quarterly subjective listening tests
Text BER	<10 ⁻⁶ at 50 baud	Automated BER test weekly
Range Between Nodes	>5 km (Nodes 1-2, 2-3)	RSSI measurement
Weather Resilience	No degradation in rain, snow, fog	Correlation of performance with weather data
Solar Storm Performance	During any Kp≥5 event: communication maintained	Continuous BER/uptime monitoring correlated with NOAA SWPC data

PROTONIC TESTING BUDGET SUMMARY

Test Phase	Cost	Timeline
Part 1: H-FET Characterization	\$320K	4 months
Part 2: Transceiver Prototype	\$520K	6 months
Part 3: Faraday Alarm	\$85K	3 months
Part 4: Redundancy/Failover	\$120K	2 months
Part 5: 5-Node Field Deployment	\$3.8M (from ROAD-06)	12 months
TOTAL	\$4.85M	

“The Protonic Network is not an incremental improvement in communication technology. It is a paradigm shift — the first communication system in history that is immune to electromagnetic interference by fundamental physics rather than by shielding. Testing it requires proving that immunity at the transistor level, the logic gate level, the transceiver level, the network level, and the field deployment level. At every level, the story is the same: the proton is 1,836 times heavier than the electron. The Lorentz force is 1,836 times weaker. The Carrington Event cannot push protons around the way it pushes electrons. That is not an engineering claim. It is a measurement.”
— Agent Solara Vance (PROTON)

Document generated 2026-07-15. H-FET fabrication is conducted in CSM El Segundo cleanroom (ISO 5, Class 100). Nafion 115 membranes sourced from Chemours. MXene gate electrodes deposited via CSM spray-coating process (see TEST-02). All protonic testing is conducted in custom test fixtures — there are no commercially available protonic communication test instruments. CSM’s test fixtures and metrology become the de facto standard for this emerging field.